Serendib University Endowment Fund: Portfolio Tear Sheet

Mandate: USD 10,000,000 Capital Pool | **Recommended Portfolio:** Maximum Sharpe | **Sample:** Nov 2004 to Oct 2026 (5,518 days)

1. Asset Allocation & Mandate Weights

Ticker	Asset Class / Fund Name	Target Weight	Capital (USD)	Mandate Limit	Status
SPY	US Large Cap Equities (S&P 500)	9.55%	$955,000	Max 40%	Pass
EFA	Developed International Equities	0.00%	$0	Max 30%	Pass
EEM	Emerging Market Equities	0.00%	$0	Max 10%	Pass
AGG	US Aggregate Fixed Income (Core Anchor)	58.85%	$5,885,000	Min 30%, Max 60%	Pass
GLD	SPDR Gold Shares (Real-Asset Diversifier)	15.00%	$1,500,000	Max 15%	Pass
XLE	Energy Select Sector	1.60%	$160,000	Max 10%	Pass
XLK	Technology Select Sector	15.00%	$1,500,000	Max 15%	Pass
Total	**Serendib Endowment Portfolio**	**100.00%**	**$10,000,000**	**Long-only, No Leverage**	**Pass**

2. Performance & Risk-Adjusted Summary

- **Annual Return: 7.4% a year historically**, in-sample. It clears our 7.0% base objective (4.5% long-run spending + 2.5% inflation), matches the stricter 7.4% hurdle (4.9% spending + 2.5% inflation), and falls short of the 7.8% hurdle (4.5% spending + 3.3% higher-education inflation).
- **Sharpe Ratio: 0.79** (Highest risk-adjusted return across all candidate portfolios).
- **Annual Volatility: 7.2%** (Conservative risk profile dominated by 58.85% core fixed income).
- **Maximum Drawdown: -18.3%** (Comparable to Minimum Variance at -17.8%, while earning about 0.9 points more a year).

3. Tail Risk Metrics (1-Day Horizon on USD 10M Capital)

Risk Framework	95% 1-Day VaR	95% 1-Day ES	99% 1-Day VaR	99% 1-Day ES	Tail Gap (ES - VaR)
Parametric Normal (Gaussian)	0.71% ($71,300)	0.90% ($90,100)	1.02% ($102,000)	1.17% ($117,300)	0.15% ($15,300)
Historical Simulation	0.66% ($66,200)	1.02% ($102,300)	1.16% ($116,300)	1.73% ($173,400)	0.57% ($57,100)
Parametric Student-t	0.61% ($60,600)	0.98% ($98,100)	**1.16% ($115,700)**	**1.73% ($172,600)**	0.57% ($56,900)
Extreme Value Theory (EVT GPD)	0.65% ($64,637)	1.02% ($102,200)	**1.22% ($122,044)**	**1.73% ($172,578)**	0.51% ($50,534)

Observation: Normal models underestimate deep tail risk by $55k+ at 99% ES. Student-t and EVT pass 99% coverage, but all models fail breach independence because losses cluster in crises.

4. Stress Testing & Crisis Scenario Losses

Scenario	Nature	Period / Assumption	Portfolio Loss (%)	Capital Impact (USD)
2008 Global Financial Crisis	Historical Replay	2008-09 to 2009-03	-5.2%	-$520,000
March 2020 COVID-19 Shock	Historical Replay	2020-02 to 2020-03	-9.9%	-$990,000
2022 Joint Rate Hiking Cycle	Historical Replay	2022-01 to 2022-12	**-12.6%**	**-$1,260,000**
Stagflationary Inflation Shock	Hypothetical (assumption)	SPY/EFA -25%, EEM -30%, AGG -15%, GLD -5%, XLE 0%, XLK -35%	**-17.2%**	**-$1,720,000**

Worst Historical Replay: 2022 rate hikes (-12.6%), driven by simultaneous declines in equities and fixed income.

5. Risk Policy Limits & Committee Governance

Governing Policy Limit	Mandated Limit Level	Final Portfolio Level	Compliance Status
One-Day 99% Value at Risk (VaR)	USD 150,000 (1.50%)	**USD 116,000 (1.16%)**	**Within limit**
One-Day 99% Expected Shortfall (ES)	USD 200,000 (2.00%)	**USD 173,000 (1.73%)**	**Within limit**
Loss over one crisis replay	USD 1,500,000 (15.00%)	**12.6% (USD 1,260,000), 2022**	**Within limit**
Inflation shock (assumption)	USD 1,500,000 (15.00%)	**17.2% (USD 1,720,000)**	**Exceeds, disclosed**

The limit is measured over the whole crisis window. Inside 2022 the fall from peak to trough reached 17.2%.

Committee Action Protocol:

- If persistent inflation materializes, rotate capital from XLK into XLE (each point shifted lowers the inflation shock loss by about 0.35 points; about 6.5 points brings it inside 15%).